

Name _____

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KNOWLEDGE:

1. Determine the derivative of each of the following functions:

a) Find the 1st and 2nd derivative for this question:

$$f(x) = 6xe^{2x-1} \quad (3)$$

$$f'(x) = 6e^{2x-1} + 6xe^{2x-1}(2)$$

$$= 12e^{2x-1}(1+2x)$$

$$f''(x) = 12e^{2x-1} + 12e^{2x-1} + 12xe^{2x-1}(2)$$

$$f''(x) = 24e^{2x-1} + 24xe^{2x-1}$$

$$f''(x) = 24e^{2x-1}(1+x)$$

$$b) y = (5^{x^2+1} - 3x^4)^3 \quad (3)$$

$$y' = 3(5^{x^2+1} - 3x^4)^2(5^{x^2+1} \ln 5(2x) - 12x^3)$$

$$y' = 6(5^{x^2+1} - 3x^4)^2(5^{x^2+1} \ln 5 - 6x^3)$$

$$c) f(x) = \frac{e^{4x} + 1}{e^{x^2}} \quad (3)$$

$$f'(x) = \frac{4e^{4x}e^{x^2} - (e^{4x} + 1)(2xe^{x^2})}{(e^{x^2})^2}$$

$$= \frac{2e^{x^2}(2e^{4x} - (e^{4x} + 1)x)}{(e^{x^2})^2}$$

$$= \frac{2(2e^{4x} - xe^{4x} - x)}{e^{x^2}}$$

2. Determine the point(s), correct to 2 decimal places, when the tangent to the function $g(x) = e^{-x}(x^2 - 3x + 1)$ is horizontal.

$$g'(x) = -e^{-x}(x^2 - 3x + 1) + e^{-x}(2x - 3)$$

$$g'(x) = -e^{-x}[(x^2 - 3x + 1) - (2x - 3)]$$

$$= -e^{-x}(x^2 - 5x + 4)$$

$$0 = -e^{-x}(x - 4)(x - 1)$$

$$x = 4, 1$$

$$g(4) = e^{-4}(4^2 - 3(4) + 1)$$

$$= e^{-4}(5)$$

$$= 0.6916$$

$$g(1) = e^{-1}(1^2 - 3(1) + 1)$$

$$= e^{-1}(-1)$$

$$= -0.3679$$

$$A(4, 0.69)$$

$$B(1, -0.37)$$

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APPLICATION:

1. The concentration of a medication in the bloodstream t hours after injection is $c(t) = 3t^2e^{-2t}$.

What is the maximum concentration within the first four hours? Provide an exact final answer. (4)

$$c(t) = 3t^2e^{-2t}$$

$$[0, 4]$$

$$c'(t) = 6te^{-2t} + 3t^2e^{-2t}(-2)$$

$$= 6te^{-2t} - 6t^2e^{-2t}$$

$$= 6te^{-2t}(1-t)$$

$$t = 0, 1$$

$$c(0) = 0$$

$$c(1) = 3e^{-2}$$

$$3e^{-2} > 48e^{-3}$$

$$c(4) = 48e^{-8}$$

→ Therefore, max concentration is $3e^{-2}$ at 1 hour.

2. The growth in the population of a group of rabbits is given by $P(t) = 800e^{0.08t}$, where P is the population at time t measured in weeks.

- a. What is the initial population of rabbits?

$$P(0) = 800e^0$$

$$= 800 \text{ rabbits}$$

- b. How many rabbits are there after 14 days?

$$P(14) = 800e^{0.08 \cdot 14}$$

$$= 800e^{1.12}$$

$$= 2452 \text{ rabbits}$$

- c. What is the rate of change of rabbits on the 21st day?

$$P'(t) = 800e^{0.08t}(0.08)$$

$$= 64e^{0.08t}$$

$$P'(21) = 64e^{0.08 \cdot 21}$$

$$= 64e^{1.68}$$

$$= 343 \text{ rabbits per week}$$